

5. (a) (i) Let T be the tension.

$$2T \cos 75^\circ = 60$$

$$T = 115.9 \text{ N}$$

1M

1A 2

- (ii) energy stored in the string = k.e. of arrow

$$= \frac{1}{2}(0.2)(45)^2$$

$$= 202.5 \text{ J}$$

1M

1A 2

- (b) (i) $d = v \cos 20^\circ t$

$$60 = 45 \cos 20^\circ t$$

$$t = 1.42 \text{ s}$$

1M

1A 2

- (ii) $h = 25 - \frac{1}{2}gt^2$

$$= 25 - \frac{1}{2}(9.81)(1.42)^2$$

$$= 15.1 \text{ m [or } h = 14.9 \text{ m]}$$

1M

1A 2